

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**  
**Semester Examination – MAY - 2019**

**Branch: SY Mechanical Engineering**  
**Subject with Subject Code: SOM (BTMEC403)**  
**Date: 20-05-2019**

**Sem.: IV**  
**Marks: 60**  
**Time: 3 Hr.**

**Instructions to the Students**

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

**(Marks)**

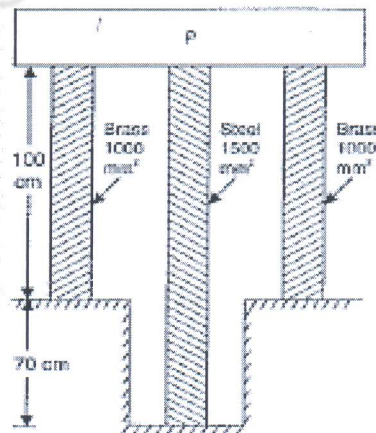
**Q.1. a) Define the following terms**

- i) Young's Modulus
- ii) Modulus of Rigidity
- iii) Poisson's ratio
- iv) Factor of safety
- v) Hook's law

**(05)**

**b) Two brass rods and one steel rod together support a load as shown in Fig below. If the stresses in brass and steel are not to exceed  $60 \text{ N/mm}^2$  and  $120 \text{ N/mm}^2$ , find the safe load that can be supported. Take  $E$  for steel =  $2 \times 10^5 \text{ N/mm}^2$  and for brass =  $1 \times 10^5 \text{ N/mm}^2$ . The cross sectional area of steel rod is  $1500 \text{ mm}^2$  and of each brass rod is  $1000 \text{ mm}^2$ .**

**(07)**



**OR**

**b) A steel rod of 3 cm diameter is enclosed centrally in a hollow copper tube of external diameter 5 cm and internal diameter of 4 cm. The composite bar is then subjected to an axial pull of 45000 N. If the length of each bar is equal to 15 cm. Determine:**

- i) The stresses in the rod and tube.  
 ii) Load carried by each bar.

(07)

**Q.2. a)** Prove that stress induced in a body when the load is applied with the impact is given by

$$\sigma = \frac{P}{A} \left( 1 + \sqrt{1 + \frac{2AEh}{P.L}} \right)$$

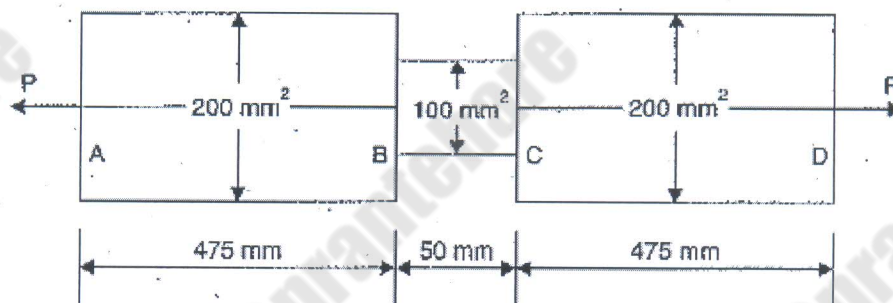
where P = Load applied with impact, A = Cross-sectional area of the body,  
 h = height through which load falls, L = Length of the body, and  
 E = Modulus of elasticity.

(06)

OR

**a)** The maximum stress produced by a pull in a bar of length 1 m is 150 N/mm<sup>2</sup>. The area of cross-sections and length are shown in Fig. Calculate the strain energy stored in the bar if E = 2 x 10<sup>5</sup> N/mm<sup>2</sup>.

(06)



**b)** The stresses at a point in a bar are 200 N/mm<sup>2</sup> (tensile) and 100 N/mm<sup>2</sup> (compressive). Determine the resultant stress in magnitude and direction on a plane inclined at 60° to the axis of the major stress. Also determine the maximum intensity of shear stress in the material at the point.

(06)

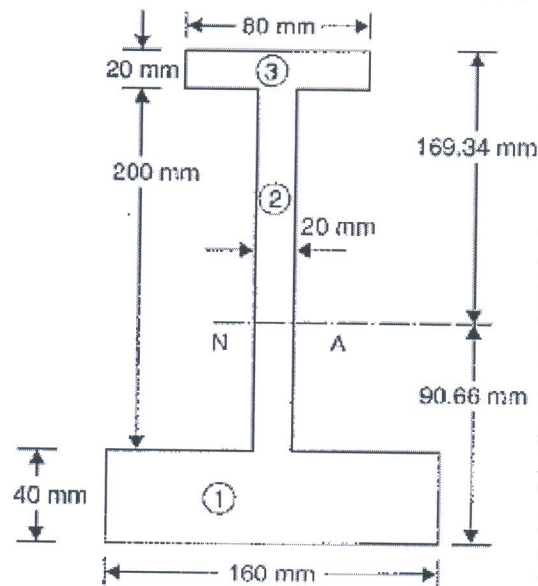
**Q.3. a)** A hollow rectangular section is having external size 500 mm x 450 mm internal size 400 mm x 350 mm. It carries a vertical load of 100 kN at outer edge of the column on X-axis. Calculate maximum and minimum intensities of stresses in the section. Assume 500 mm side horizontal.

(06)

**b)** Draw the shear force and bending moment (B.M.) diagram for a simply supported beam of length 9 m and carrying a uniformly distributed load of 10 kN/m for a distance of 6 m from the left end. Also calculate the maximum B.M. on the section.

(06)

- Q.4.** A cast iron beam is of I-section as shown in Fig. The beam is simply supported on a span of 5 m. If the tensile stress is not to exceed  $20 \text{ N/mm}^2$ , Find the safe uniformly load which the beam can carry. Find also the maximum compressive stress. (12)



- Q.5. a)** A cantilever of length 3 m carries a point load of 10 kN at a distance of 2 m from the fixed end. If  $E = 2 \times 10^5 \text{ N/mm}^2$  and  $I = 10^8 \text{ mm}^4$ , find the slope and deflection at the free end using conjugate beam method. (05)

OR

- a)** A cantilever of length 2 m carries a point load of 20 kN at the free end and another load of 20 kN at its center. If  $E = 10^5 \text{ N/mm}^2$  and  $I = 10^8 \text{ mm}^4$  for the cantilever then determine by moment area method, the slope and deflection of the cantilever at the free end. (05)

- b)** A cantilever of length 2 m carries a uniformly distributed load  $2 \text{ kN/m}$  over a length of 1 m from the free end, and a point load of 1 kN at the free end. Find the slope and deflection at the free end if  $E = 2.1 \times 10^5 \text{ N/mm}^2$  and  $I = 6.667 \times 10^7 \text{ mm}^4$ . (07)

- Q.6. a)** Derive the relation for a circular shaft when subjected to torsion as given below

$$\frac{T}{J} = \frac{\tau}{R} = \frac{C\theta}{L}$$

where  $T$  = Torque transmitted,

$J$  = Polar moment of inertia,  
 $\tau$  = Max. Shear stress,  
 $R$  = Radius of the shaft,  
 $C$  = Modulus rigidity,  
 $\theta$  = Angle of twist, and  
 $L$  = Length of the shaft.

(06)

b) A hollow shaft, having an inside diameter 60% of its outer diameter, is to replace a solid shaft transmitting the same power at the same speed. Calculate the percentage saving in material, if the material to be used is also the same.

(06)

OR

b) A solid round bar 4 m long and 5 cm in diameter was found to extend 4.6 mm under a tensile load of 50 kN. This bar is used as a strut with both ends hinged. Determine the buckling load for the bar and also the safe load taking factor of safety as 4.

(06)

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,**  
**LONERE – RAIGAD -402 103**  
**Supplementary Winter Semester Examination: Nov. – 2019**

**Branch:** Mechanical Engineering  
**Subject:** - Strength of Materials (BTMEC403)  
**Date:-** 30/11/2019

**Sem.:- IV**  
**Marks: 60**  
**Time:- 3 Hr.**

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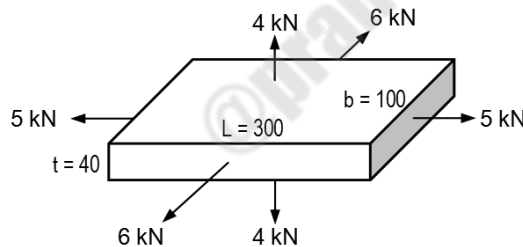
Que.1 a) State and explain following terms

- i Hooke's law
- ii Bulk Modulus
- iii Poisson's ratio

**6 Marks**

- b) A metallic bar  $300 \text{ mm} \times 100 \text{ mm} \times 40 \text{ mm}$  is subjected to external forces as shown in fig. 1. Determine change in volume of the block. Take  $E = 2 \times 10^5 \text{ N/mm}^2$ ,  $\mu = 0.25$ .

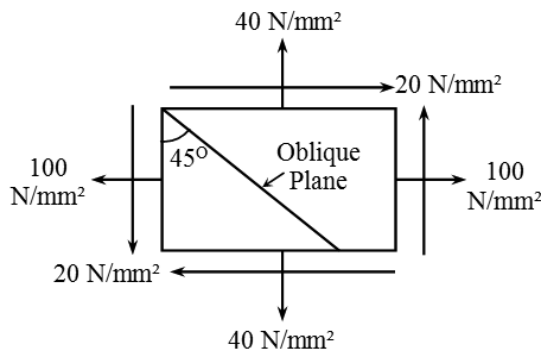
**6 Marks**



**Figure.1**

- Que.2 a) A point in a strained material is subjected to stresses shown in fig. 2. Determine normal, tangential and resultant stresses across oblique plane by Mohr's circle method. Also calculate angle of obliquity.

**8 Marks**



**Figure.2**

- b) A bar 1.5 m long and 10 mm diameter hangs vertically and has a collar securely fixed at the lower end. Find the maximum stress induced in the bar when a load of 150 N falls on the collar from a height of 25 mm. Take  $E = 2 \times 10^5 \text{ N/mm}^2$ . Also find strain energy stored in the bar. **4 Marks**

- Que.3 a) A square column 300 mm  $\times$  300 mm carries an axial load of 200 kN. Find the position of 30 kN load along the axis bisecting the width of cross section so that end stresses developed at the other extreme of the column will be zero. **6 Marks**

- b) A simply supported beam AB of span 4 m carries an uniformly distributed load as shown in fig. 3. Draw S.F. and B.M. diagrams for the beam. Find magnitude of maximum B.M. **6 Marks**

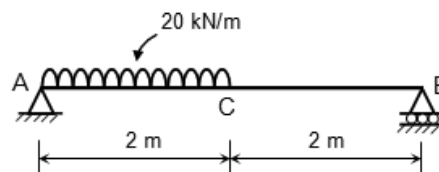


Figure.3

- Que.4 a) A rectangular beam 300 mm wide and 200 mm deep is simply supported over a span of 8 m. What u.d.l. per meter the beam may carry, if the bending stress is not to exceed  $120 \text{ N/mm}^2$ . **4 Marks**

- b) A channel section shown in fig. 5 (a) is used as a beam loaded as shown in fig. 5 (b). Draw the shear stress distribution diagram for the cross-section carrying maximum shear force. **8 Marks**

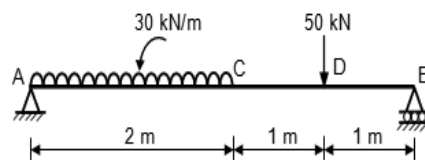


Fig. 5 (a)

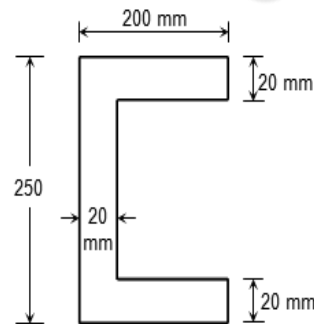
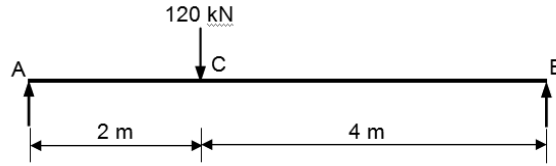


Fig. 5 (b)

Que.5 a) A simply supported beam carrying a point load is shown in fig.6. Determine **8 Marks**

- 1) Slope at A and B
- 2) Maximum deflection.

Take  $E = 200 \text{ GPa}$ ,  $I = 60 \times 10^6 \text{ mm}^4$ .



**Figure.6**

b) A simply supported beam 5 meter long, carries 10 kN of load at the mid-point. Calculate by moment area method, the slope at the supports and deflection at the mid-span. **4 Marks**

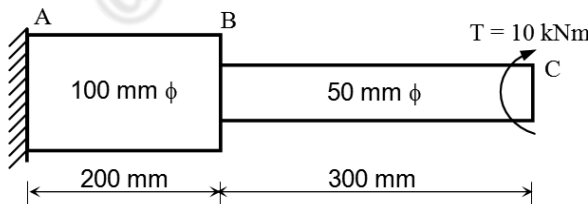
Take  $E = 2 \times 10^5 \text{ N/mm}^2$  and  $I = 1 \times 10^6 \text{ mm}^4$ .

Que.6. a) Two shaft AB and BC are connected in series as shown in fig. 7. The **6 Marks**

diameters of AB and BC are 100 mm and 50 mm respectively and their lengths are 200 mm and 300 mm respectively. Both the shafts are made of the same material having modulus of rigidity as  $8 \times 10^4 \text{ N/mm}^2$ . Determine :

- i) Shear stresses set up in each shaft, and
- ii) The total angle of twist.

The torque applied at the one end is 10 kNm.



**Figure.7**

b) The external and internal diameter of a hollow C.I. column is 5 cm and 4 cm respectively. If the length of its column is 3 m and both of its ends are fixed. Determine the crippling load using Rankine's formula. **6 Marks**

Take  $\sigma_c = 550 \text{ N/mm}^2$  and  $\alpha = 1/1600$  in Rankine's formula.

----- PAPER END -----

**Instructions to the Students:**

1. All the questions are compulsory
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question
3. Use of non-programmable scientific calculators is allowed
4. Assume suitable data wherever necessary and mention it clearly

(Level/CO) Marks

**Q.1 Attempt Any Two**

- A) Derive an expression for deformation of uniformly tapering circular cross-sectional body (CO1) 6
- B) A composite bar made of copper, steel and brass is rigidly attached to the end supports as shown in figure 01. Determine the stresses in the three portions of the bar when the temperature of the composite system is raised by  $70^{\circ}\text{C}$  when  
i) The supports are rigid ii) the supports yield by 0.6 mm.  
 $E_c = 100 \text{ GPa}$ ,  $E_s = 205 \text{ GPa}$ ,  $E_b = 95 \text{ GPa}$   
 $\alpha_c = 18 \times 10^{-6} / ^{\circ}\text{C}$ ,  $\alpha_s = 11 \times 10^{-6} / ^{\circ}\text{C}$ ,  
 $\alpha_b = 19 \times 10^{-6} / ^{\circ}\text{C}$

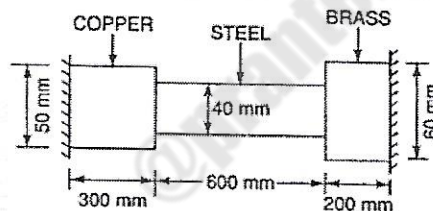


Figure 01

- C) A plane element in a boiler is subjected to tensile stresses of 400 MPa on one plane and 150 MPa on the other at right angle. Each of the above stresses is accomplished by a shear stress of 100 MPa such that when associated with the minor stress tends to rotate the element in anticlockwise direction. Find;  
i. Principal stresses and their direction  
ii. Maximum shear stress (CO3) 6

**Q.2 Attempt Any Two**

- A) A wagon weighing 20 kN is attached to a wire rope and is moving at the speed of 5.4 kmph. The rope suddenly jams and wagon is brought to rest. If length of rope is 50 m and diameter is 36 mm, find maximum instantaneous stress and elongation of rope. Take  $E = 200 \text{ GPa}$ . (CO2) 6
- B) A rectangular pier is subjected to a compressive load of 450 kN as shown in figure 02. Find stress intensities on all the four corners of the pier.

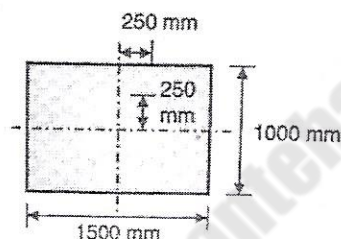


Figure 02

- C) Interpret a relationship between rate of loading, shear force and bending moment (CO4) 6

**Q.3 Attempt Any Two**

- A) An I-section beam 350 mm X 150 mm has a web thickness 10 mm & flange thickness 20 mm. If the shear force acting on the section is 40 kN, then find;  
 a) Maximum shear stress developed in the section (CO3) 6  
 b) Sketch the shear stress distribution diagram  
 c) Total shear force carried by web
- B) Derive an expression for Flexural equation along with assumptions (CO2) 6
- C) A beam of T-section, 4 m long carries a uniformly distributed load 'w' per meter run throughout its length. The beam is simply supported at its ends. The T-section has web 18.8 cm X 1.2 cm and flange is 10 cm X 1.2 cm. What is the maximum value of 'w', so that the stress in the section does not exceed 60 MPa? (CO3) 6

**Q.4 Attempt Any Two**

- A) Derive an expression for Torsional formula along with assumptions (Understand) 6
- B) A Hollow shaft with diameter ratio of 3/8 is required to transmit 500 kW at 100 rpm, the maximum torque being 20% greater than mean. The maximum shear stress is not to exceed 60 N/mm<sup>2</sup> and the twist in the length of 3 m is not to exceed 1.4°. Calculate the minimum diameter required for the shaft. (Apply) 6  
 Take  $G = 84 \text{ N/mm}^2$ .
- C) A hollow CI column of external diameter 200 mm, length 4 meter with both the ends fixed, supports an axial load of 800 kN. Determine the thickness of the column required by using Rankine's formula taking constant of 1/6400 & working stress at 80 MN/m<sup>2</sup> (Apply) 6

**Q.5 Attempt Any Two**

- A) Draw shear force and bending moment diagrams for the beam loaded as shown in figure 03

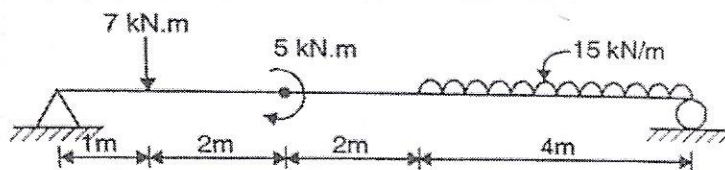


Figure 03

- B) Draw shear force and bending moment diagrams for the cantilever beam as shown in figure 04

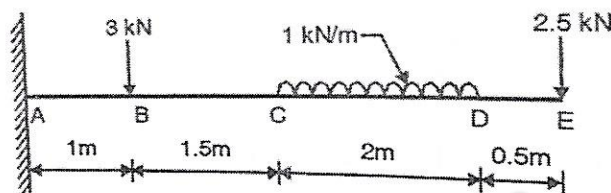


Figure 04

- C) A beam ABCD has an internal hinge at B and is loaded shown in figure 05. Plot shear force and bending moment diagrams and locate point of contra flexure. (CO4) 6

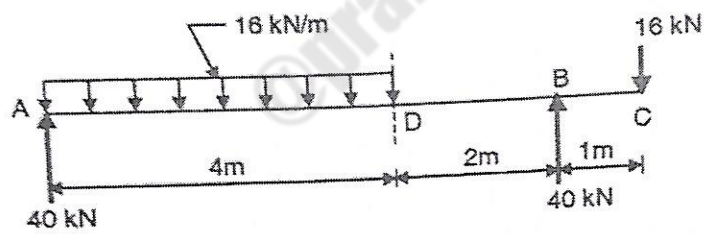


Figure 05

\*\*\* END \*\*\*

|                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                         |            |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|
| <b>DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE</b><br><b>Supplementary Semester Examination – Summer 2022</b><br><b>Course: B. Tech.                      Branch : Mechanical Engg.                      Semester : IV</b><br><b>Subject Code &amp; Name: BTMEC403 – Strength of Materials</b><br><b>Max Marks: 60                      Date:                      Duration: 3.00 Hrs.</b> |                                                                                                                                                                                                                                                                                                                                                                                                                         |            |       |
| <b>Instructions to the Students:</b><br>1. All the questions are compulsory.<br>2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.<br>3. Use of non-programmable scientific calculators is allowed.<br>4. Assume suitable data wherever necessary and mention it clearly.                     |                                                                                                                                                                                                                                                                                                                                                                                                                         |            |       |
|                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                         | (Level/CO) | Marks |
| <b>Q. 1</b>                                                                                                                                                                                                                                                                                                                                                                                            | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                  |            |       |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                                              | Define and explain the term ‘Poisson’s ratio’. How is it calculated based on lateral and axial strains?                                                                                                                                                                                                                                                                                                                 | CO1        | 06    |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                                                                              | A short hollow circular tube (with vertical axis) is applied with an axial load of 5 kN. The inner and outer diameters of the tube are 75 mm and 100 mm, respectively. The original length of the tube is 500 mm which is reduced by 0.2 mm after applying the load. Determine the compressive stress and strain in the tube. (Neglect the self weight of the tube and assume the tube does not buckle under the load). | CO2        | 06    |
| <b>C)</b>                                                                                                                                                                                                                                                                                                                                                                                              | A stepped metal rod ABC (as shown in figure below) is subjected to an axial pull of 7 kN. The diameters of the two sections AB and section BC of the rod are 65 mm and 30 mm respectively. Find out in which of the two sections the stress will be maximum? Also find out the stresses in section AB and BC of the rod. <div data-bbox="370 1422 1056 1664" data-label="Image"> </div>                                 | CO2        | 06    |
|                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                         |            |       |
| <b>Q.2</b>                                                                                                                                                                                                                                                                                                                                                                                             | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                  |            |       |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                                              | Three steel plates of 18 mm thickness each are joined with the help of two rivets of 15 mm diameter each. If the plates are pulled (as shown in the figure below) with $P = 6$ kN, calculate the maximum shear stress induced in the rivets.                                                                                                                                                                            | CO3        | 06    |

|                                             |                                                                                                                                                                                                                                                |      |    |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----|
|                                             |                                                                                                                                                                                                                                                |      |    |
| B)                                          | The tensile stresses at a point across mutually perpendicular planes are $120 \text{ N/mm}^2$ and $60 \text{ N/mm}^2$ . Determine the normal, tangential and resultant stresses on a plane inclined at $30^\circ$ to the axis of minor stress. | CO3  | 06 |
| C)                                          | Define and explain following terms:<br>A. Resilience<br>B. Proof Resilience<br>C. Modulus of Resilience                                                                                                                                        | CO3  | 06 |
|                                             |                                                                                                                                                                                                                                                |      |    |
| <b>Q. 3 Solve Any Two of the following.</b> |                                                                                                                                                                                                                                                |      |    |
| A)                                          | For the cantilever beam as shown in figure below, calculate the shear force and bending moment and draw the shear force and bending moment diagram.                                                                                            | CO04 | 06 |
| B)                                          | A simply supported beam 4 m span is carrying loads as shown in figure. Draw shear force and bending moment diagrams for the beam.                                                                                                              | CO4  | 06 |
| C)                                          | Describe various types of supports for beam with the help of neat sketch.                                                                                                                                                                      | CO4  | 06 |
|                                             |                                                                                                                                                                                                                                                |      |    |
| <b>Q.4 Solve Any Two of the following.</b>  |                                                                                                                                                                                                                                                |      |    |
| A)                                          | A wooden beam 100 mm wide, 250 mm deep and 3 m long is carrying a uniformly distributed load of $40 \text{ kN/m}$ . Determine the maximum shear stress                                                                                         |      | 06 |

|             |                                                                                                                                                                                                                                                                                                                                              |     |    |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|
|             | and sketch the variation of shear stress and along the depth of the beam.                                                                                                                                                                                                                                                                    |     |    |
|             |                                                                                                                                                                                                                                                                                                                                              |     |    |
| <b>B)</b>   | Explain the assumptions of theory of simple bending.                                                                                                                                                                                                                                                                                         |     | 06 |
| <b>C)</b>   | A beam 6 m long, simply supported at its ends, is carrying a point load of 50 kN at its center. The moment of inertia of the beam is given as equal to $78 \times 10^6 \text{ mm}^4$ . If E for the material of the beam is $2.1 \times 10^5 \text{ mm}^4$ . Calculate: 1. Deflection at the center of the beam and 2. Slope at the support. | CO5 | 06 |
|             |                                                                                                                                                                                                                                                                                                                                              |     |    |
| <b>Q. 5</b> | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                       |     |    |
| <b>A)</b>   | Find the maximum shear stress induced in a solid circular shaft of diameter 15 cm when the shaft transmits 150 kW power at 180 rpm.                                                                                                                                                                                                          | CO5 | 06 |
|             |                                                                                                                                                                                                                                                                                                                                              |     |    |
| <b>B)</b>   | Describe various types of end conditions of columns.                                                                                                                                                                                                                                                                                         | CO6 | 06 |
|             |                                                                                                                                                                                                                                                                                                                                              |     |    |
| <b>C)</b>   | Write short note on 'Limitation of Euler's formula'.                                                                                                                                                                                                                                                                                         | CO6 | 06 |
|             | <b>*** End ***</b>                                                                                                                                                                                                                                                                                                                           |     |    |

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(Level Marks  
/CO)

**Q. 1 Solve Any Two of the following.**

**12**

- A) Derive from the fundamental, the relation for the deformation of a body, when it is subjected to
- a) A tensile force
  - b) Its own weight.
- R/CO-1 6
- B) A steel bar of 600 mm<sup>2</sup> cross sectional area is carrying loads as shown in fig. No. 1 U/CO-1 6

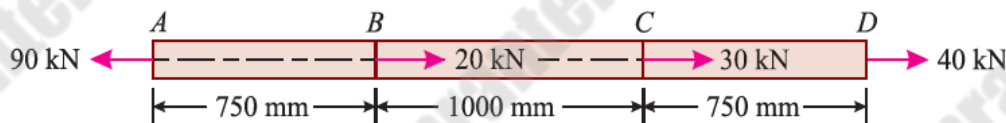


Fig. No. 1

- C) An element in strained body is subjected to a tensile stress of 150 MPa and shear stress of 50 Mpa tending to rotate the element in and anticlockwise direction. Find (i) the magnitude of the normal and shear stresses on a section inclined at 40° with the tensile stress and (ii) The magnitude and direction of maximum shear stress that can exist on the element. A/CO-1 6

**Q.2 Solve Any Two of the following.**

**12**

- A) A vertical round steel rod, 1.82 m long is securely held at its upper end and a weight sliding freely on the rod falls on to a stop at the lower end of the rod. When the weight falls from a height of 30 mm the maximum stress is reached in the rod is estimated to be 157 N/mm<sup>2</sup>. Determine the stress if the load had been gradually applied and also the maximum stress if the load has fallen from a height of 45 mm. take  $E = 2 \times 10^5 \text{ N/mm}^2$ . U/CO-2 6
- B) Derive the equation for Strain energy for
- i) Gradually applied load
  - ii) Suddenly Applied load.
- R/CO-2 6

- C) A rectangular strut 200 mm wide and 150 mm thick carries a load of 60 kN at an eccentricity of 20 mm in a plane bisecting the thickness. Find the maximum and minimum intensities of stress in the section. Also draw stress distribution diagram. A/CO-3 6

**Q. 3 Solve Any Two of the following.** 12

- A) State the assumptions made in the theory of pure bending. R/CO-4 6
- B) An I sections with rectangular ends, has the following dimensions: Flanges 150 mm x 20 mm, web 300 mm x 10 mm. Find the maximum shearing stress developed in the beam for a shear force of 50 kN. U/CO-4 6
- C) A solid steel shaft has to transmit 100 kW at 160 r.p.m. taking allowable shear stress as 70 Mpa, find the suitable diameter of the shaft. The maximum torque transmitted in each revolution exceeds the mean by 20 %. A/CO-4 6

**Q.4 Solve Any Two of the following.** 12

- A) State and explain the relation between  
i) The load and Shear force  
ii) The Shear force and Bending moment. R/CO-5 6
- B) A simply supported beam of span 3 m carries two-point loads of 5 kN each at 1 m and 2 m from the left-hand support. Draw shear force and bending moment diagrams for the beam. U/CO-5 6
- C) The intensity of loading on a simply supported beam of 6 m span increases gradually from 800 N/m run at one end to 2000 N/m run at another end as shown in fig. No. 2. Find the position and amount of maximum bending moment. Also draw the shear force and bending moment diagram. A/CO-5 6

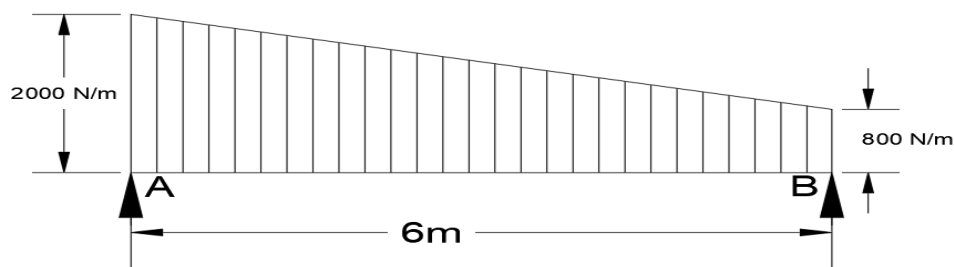


Fig. No. 2

**Q. 5 Solve Any Two of the following.** 12

- A) Derive the equation for the slope and deflection for the beam subjected to pure bending. R/CO-6 6

- B) A simply supported beam AB of Span 5 m is carrying a point load of 30 kN at a distance 3.75 m from the left end A. Calculate the slopes at A and B and deflection under the load. Take  $EI = 26 \times 10^7 \text{ N-mm}^2$ . U/CO-6 6
- C) Determine slope and deflection at the free end and at point C, 2 m from fixed end A for the cantilever beam shown in fig. No. 3 using moment area method. A/CO-6 6

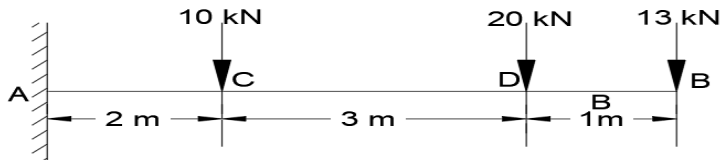


Fig. No. 3

\*\*\* End \*\*\*

**Instructions to the Students:**

1. All the questions are compulsory
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question
3. Use of non-programmable scientific calculators is allowed
4. Assume suitable data wherever necessary and mention it clearly

(Level/CO) Marks

**Q.1 Attempt Any Two**

- A) Draw stress-strain curve for mild steel. Explain characteristic points of the curve. (CO1) 6
- B) A rod of steel is 10 meters long at a temperature of 20 degrees Celsius. Find the free expansion of the rod when the temperature is raised to 65 degrees Celsius. Find the temperature stress produced when the expansion of the rod is prevented. Take  $E = 200 \text{ GPa}$  and  $\alpha = 12 \times 10^{-6}$  per degree Celsius (CO2) 6
- C) A plane element in a boiler is subjected to tensile stresses of 400 MPa on one plane and 150 MPa on the other at right angle. Each of the above stresses is accomplished by a shear stress of 100 MPa such that when associated with the minor stress tends to rotate the element in anticlockwise direction. Find; (CO3) 6
- i. Principal stresses and their direction
  - ii. Maximum shear stress

**Q.2 Attempt Any Two**

- A) A wagon weighing 20 kN is attached to a wire rope and is moving at the speed of 5.4 kmph. The rope suddenly jams and wagon is brought to rest. If length of rope is 50 m and diameter is 36 mm, find maximum instantaneous stress and elongation of rope. Take  $E = 200 \text{ GPa}$ . (CO2) 6
- B) A rectangular pier is subjected to a compressive load of 450 kN as shown in figure 02. Find stress intensities on all the four corners of the pier.

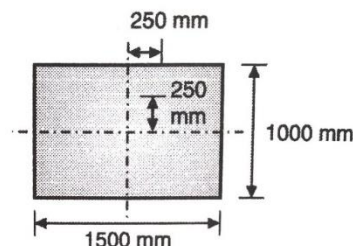


Figure 02

- C) Interpret a relationship between rate of loading, shear force and bending moment (CO4) 6

**Q.3 Attempt Any Two**

- A) An I-section beam 350 mm X 150 mm has a web thickness 10 mm & flange thickness 20 mm. If the shear force acting on the section is 40 kN, then find; (CO3) 6
- a) Maximum shear stress developed in the section
  - b) Sketch the shear stress distribution diagram
  - c) Total shear force carried by web

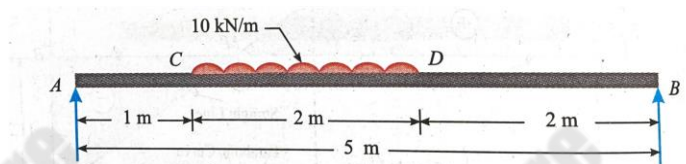
- B) Derive an expression for Flexural equation along with assumptions (CO2) 6
- C) Define the neutral axis in the context of pure bending. Discuss how it relates to the distribution of bending stress. (CO3) 6

#### Q.4 Attempt Any Two

- A) Derive an expression for Torsional formula along with assumptions (Understand) 6
- B) A hollow shaft is to transmit 200 kW at 80 r.p.m. If the shear stress is not to exceed 60 MPa and internal diameter is 0.6 of the external diameters, find the diameters of the shaft. (Apply) 6
- C) A hollow CI column of external diameter 200 mm, length 4 meter with both the ends fixed, supports an axial load of 800 kN. Determine the thickness of the column required by using Rankine's formula taking constant of  $1/6400$  & working stress at  $80 \text{ MN/m}^2$  (Apply) 6

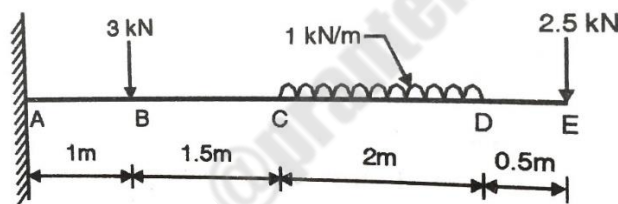
#### Q.5 Attempt Any Two

- A) Draw shear force and bending moment diagrams for the beam loaded as shown in figure below



(CO4) 6

- B) Draw shear force and bending moment diagrams for the cantilever beam as shown in figure 04



(CO4) 6

Figure 04

- C) Draw the shear force diagram and bending moment diagram for a cantilever beam carrying a point load at its free end. (CO4) 6

\*\*\* Best of Luck \*\*\*

**Instructions to the Students:**

1. All the questions are compulsory
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3. Use of non-programmable scientific calculators is allowed
4. Assume suitable data wherever necessary and mention it clearly

(Level/CO) Marks

**Q.1 Attempt Any Two**

A) Describe the stress-strain curve for a typical ductile material. Include the major regions of the curve and explain the behavior of the material in each region. (CO1) 6

B) A composite bar made of copper, steel and brass is rigidly attached to the end supports as shown in figure 01. Determine the stresses in the three portions of the bar when the temperature of the composite system is raised by  $70^{\circ}\text{C}$  when

i) The supports are rigid ii) the supports yield by 0.6 mm.

$E_c = 100 \text{ GPa}$ ,  $E_s = 205 \text{ GPa}$ ,  $E_b = 95 \text{ GPa}$

$\alpha_c = 18 \times 10^{-6} / ^{\circ}\text{C}$ ,  $\alpha_s = 11 \times 10^{-6} / ^{\circ}\text{C}$ ,

$\alpha_b = 19 \times 10^{-6} / ^{\circ}\text{C}$

(CO2) 6

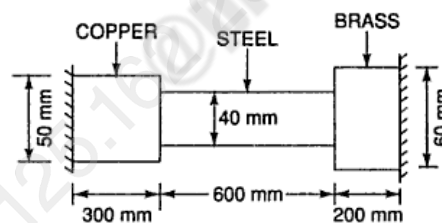


Figure 01

C) From the state of stress at a point in a strained material, it is found that  $\sigma_x = 200 \text{ MPa}$  (Tensile);  $\sigma_y = 150 \text{ MPa}$  (Compressive) and  $\tau_{xy} = 120 \text{ MPa}$ . Determine analytically,

(CO3) 6

- (i) Principal stresses and direction of principal planes
- (ii) Maximum shear stress
- (iii) Normal stress on plane of maximum shear stress.

**Q.2 Attempt Any Two**

- A)** A wagon weighing 20 kN is attached to a wire rope and is moving at the speed of 5.4 kmph. The rope suddenly jams and wagon is brought to rest. If length of rope is 50 m and diameter is 36 mm, find maximum instantaneous stress and elongation of rope. Take  $E = 200 \text{ GPa}$ . (CO2) 6
- B)** Define short and long columns and discuss their behavior under compressive loads. Explain Euler's and Rankine's formulae for calculating the critical buckling load of a column. (CO3) 6
- C)** A steel beam with a cross-sectional area of  $3000 \text{ mm}^2$  and a length of 4 meters is subjected to an axial load of 20 kN and a bending moment of  $40 \text{ kN}\cdot\text{m}$ . Calculate the maximum permissible axial load and bending moment if the middle third rule is to be satisfied. (CO4) 6

### Q. 3 Attempt Any Two

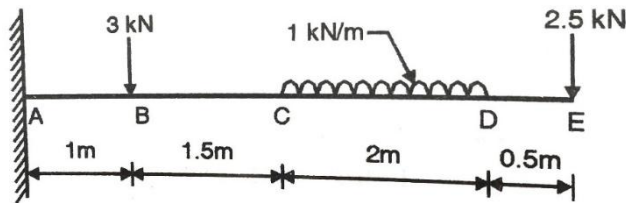
- A)** An I-section beam 350 mm X 150 mm has a web thickness 10 mm & flange thickness 20 mm. If the shear force acting on the section is 40 kN, then find;  
 a) Maximum shear stress developed in the section (CO3) 6  
 b) Sketch the shear stress distribution diagram  
 c) Total shear force carried by web
- B)** Derive an expression for Flexural equation along with assumptions (CO3) 6
- C)** A Hollow shaft with diameter ratio of  $\frac{3}{8}$  is required to transmit 500 kW at 100 rpm, the maximum torque being 20% greater than mean. The maximum shear stress is not to exceed  $60 \text{ N/mm}^2$  and the twist in the length of 3 m is not to exceed  $1.4^\circ$ . Calculate the minimum diameter required for the shaft. (CO4) 6
- Take  $G = 84 \text{ N/mm}^2$ .

### Q.4 Attempt Any Two

- A)** Draw shear force diagram (SFD) and Bending moment diagram (BMD) for a simply supported beam of length  $L = 8 \text{ m}$  carrying vertically downward point loads of 4 kN, 10 kN, and 7 kN at a distance of 1.5 m, 4 m, and 6 m respectively from left hand support. Also determine the maximum bending moment developed in the (CO5) 6

beam.

- B) Draw shear force and bending moment diagrams for the cantilever beam as shown in figure 02

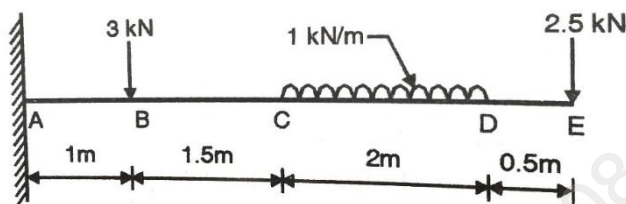


(CO5)

6

Figure 02

- C) Draw shear force and bending moment diagrams for the cantilever beam as shown in figure 03



(CO5)

6

Figure 03

#### Q. 5 Attempt Any Two

- A) Explain the principle of Macaulay's method and describe the steps involved in applying Macaulay's method for calculating deflection in beams subjected to complex loading conditions. (CO6) 6
- B) A composite beam consists of a steel section and a timber section, each 2 meters long. The beam is simply supported with a span of 4 meters and subjected to a triangular load varying from 0 kN at one end to 20 kN at the other end. Determine the total deflection at the midpoint of the beam using the moment-area method. Given:  $E_{\text{steel}}=200 \text{ GPa}$ ,  $E_{\text{timber}}=12 \text{ GPa}$ ,  $I_{\text{steel}}=400 \text{ mm}^4$ ,  $I_{\text{timber}}=150 \text{ mm}^4$ . (CO6) 6
- C) An overhanging beam is supported at both ends and has two unequal spans of lengths 3 meters and 4 meters, respectively. The beam is subjected to a uniformly distributed load of 15 kN/m over the entire length and a point load of 40 kN at a distance of 1 meter from the left support. Calculate the maximum deflection at the midpoint of the longer span using the double integration method. Given:  $E=210 \text{ GPa}$ ,  $I=600 \text{ mm}^4$ . (CO6) 6

\*\*\*END\*\*\*